What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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PDS and ACF are Fourier dual

- We can write power spectral density (or power density or power density spectrum), $S(\omega)$, as:

$$S(\omega) = \lim_{\tau \to \infty} \frac{|X(\omega)|^2}{\tau}$$

- And the autocorrelation function of power (or periodic) signal $x(t)$ with any time period T is given by:

$$R(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)\, x^*(t-\tau)dt$$

- Where, τ is called *the delayed parameter*.
- We want to prove that the power spectral density function $S(\omega)$ and the autocorrelation function $R(\tau)$ of a power signal form a Fourier transform pair, i.e.:

$$R(\tau) \Leftrightarrow S(\omega)$$

- The autocorrelation function of a power signal $x(t)$ in terms of exponential Fourier series coefficients is given by,

$$R(\tau) = \sum_{n=-\infty}^{+\infty} C_n C_{-n} e^{jn\omega_0\tau}$$

- Where, C_n and C_{-n} are the exponential Fourier series coefficients.
- Since $C_n C_{-n} = |C_n|^2$ we have:

$$R(\tau) = \sum_{n=-\infty}^{+\infty} |C_n|^2 e^{jn\omega_0\tau}$$

- By taking the Fourier transform on both sides we get:

$$F[R(\tau)] = F[\sum_{n=-\infty}^{+\infty} |C_n|^2 e^{jn\omega_0\tau}] = \int_{-\infty}^{+\infty} [\sum_{n=-\infty}^{+\infty} |C_n|^2 e^{jn\omega_0\tau}] e^{-j\omega\tau} d\tau$$

- By interchanging the order of integration and summation on RHS of the above expression, we also have:

$$F[R(\tau)] = \sum_{n=-\infty}^{+\infty} |C_n|^2 \int_{-\infty}^{+\infty} e^{jn\omega_0\tau} e^{-j\omega\tau} d\tau = \sum_{n=-\infty}^{+\infty} |C_n|^2 \int_{-\infty}^{+\infty} e^{-j\tau(\omega - n\omega_0)} d\tau$$

- Since $\int_{-\infty}^{+\infty} e^{-j\tau(\omega-n\omega_0)} d\tau = 2\pi\delta(\omega - n\omega_0)$, we have:

$$F[R(\tau)] = 2\pi \sum_{n=-\infty}^{+\infty} |C_n|^2 \delta(\omega - n\omega_0)$$

- The RHS is the power spectral density (PSD) of the power function $x(t)$. Therefore:

$$F[R(\tau)] = S(\omega)$$

- Hence, it proves that the autocorrelation function $R(\tau)$ and PSD function $S(\omega)$ of a power signal form the Fourier transform pair.

Credits

This notebook contains material obtained from https://www.tutorialspoint.com/power-spectral-density-psd-and-autocorrelation-function#.

Course Flow

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